Step 9 Summary and Lab Task Discussion (roughly one page, as required)

Plots are in figures/ and full numeric tables are in results_1_1.pdf; this page is the written discussion only.

Step 9 summary. Unloaded (T = 50 ms, D = 40 ms, 150 jobs), execution time in isolation averaged 15.96 ms (32% of T), with a 95th percentile of 24.89 ms and a measured maximum of 40.33 ms — a measured, not proven, WCET (Section 1.7): no campaign guarantees the worst path and hardware state were exercised. Release jitter was small but non-zero even unloaded (std 0.04 ms, max 0.55 ms), consistent with the mild OS-noise sources of Section 1.6; completion jitter was far larger (std 6.14 ms) because it inherits execution-time variability. 98.7% of jobs met D. A periodic sensor or filter update like this would realistically face a **firm** deadline in an industrial or automotive controller (Section 1.1): a late result is stale and discarded but not individually catastrophic. A hard classification is unsupported without a sound WCET bound, and soft suits multimedia-style tasks better.

Task 1 (background thread). Release-jitter std rose to 3.00 ms (max 34.35 ms), mean response rose from 16.54 to 26.00 ms, and the hit fraction fell from 98.7% to 90.0%. In CPython both threads share one interpreter lock, so the periodic loop waits for the busy thread, reproducing OS interference in mild form.

Task 2 (adversarial inputs). The worst-case input (maximum size plus the deep code path) measured a mean of 46.40 ms and a maximum of 71.11 ms, well above the ordinary maximum of 40.33 ms (1.76x). Neither figure is a sound WCET bound: both are measurements, and a longer path or worse hardware state may still exist.

Task 3 (decay rate). With decay constants of 2, 10 and 50 ms, the mean soft utility of the loaded run was 0.913, 0.950 and 0.984 — identical timing, three different consequences. The rate should be set by how quickly a late result actually loses value to the application, not chosen for convenience.

Task 4 (period sweep). Release jitter as a percentage of T was largest at T = 30 ms (2.0% std, 15.8% max, 84% hit), because D = 24 ms is often shorter than the job's execution time, forcing overruns into the next period. At T = 50, 100 and 200 ms, jitter stayed under 3% of T with 100% hit. Jitter matters even when every deadline is met, since periodic sampling assumes a constant interval.